

GOLDSCHMIDT 2025

Orbitrap-IRMS Abstracts (Posters and Oral Presentations)

MONDAY

Oral Presentations

Shawar et al., “Site-Specific Isotope Analysis Elucidates Mechanisms of Vanillin Degradation Under Biotic and Abiotic Transformations”

Lignin, Earth’s most abundant biopolymer derived from vascular plants, is crucial in terrestrial carbon cycling (Crawford, 1981). Its degradation monomers, such as vanillin, along with their isotope compositions serve as molecular markers to trace the terrestrial organic matter transformation within the carbon cycle (Edelkraut, 1996), and elucidate lignin decomposition pathways and environmental impacts (e.g., Bahri et al., 2008). However, the understanding of how isotopic compositions of vanillin at specific atomic positions are altered through its formation and degradation processes remains limited.

This research examines how biotic and abiotic degradation affects the site-specific carbon isotope composition of vanillin, employing unlabeled compounds with natural isotopic abundances. Preliminary results demonstrate that biotic degradation by *Pseudomonas putida* KT2440, a model soil bacterium (Simon et al., 2014). This form of biotic transformation resulted in an increase in the $d^{13}C$ in the carbonyl carbon in the residual vanillin, with modest changes to other carbons within the molecule. In contrast, abiotic transformation of vanillin via photodegradation also increases the $d^{13}C$ of the aldehyde group of the residue, but with a lesser amplitude than for biotic degradation, and accompanied by larger changes in the $d^{13}C$ of other carbons within the molecule.

This study underscores the efficacy of position-specific isotope analysis for reconstructing environmental degradation processes, as bulk isotope analysis tends to overlook subtle nuances in isotopic composition at the atomic-site level. Our approach of utilizing isotopes in their natural-abundance provides a variety of constraints without the need for prior introduction of labeled substrates, unlike conventional isotope labeling techniques (e.g., Bahri et al., 2008). This enhancement paves the way for future applications in understanding the persistence and transformation of lignin and its monomers within natural settings.

References

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Simon, O. et al., 2014. Comprehensive proteome analysis of the response of *Pseudomonas putida* KT2440 to the flavor compound vanillin. *J. Proteomics* 109, 212–227.

Poster Presentations

Saville et al., “Ground-truthing Electrospray-Orbitrap oxyanion isotopologue ratios: towards ice core clumped isotope palaeo-oxidation proxies?”

Modern gas-source mass spectrometers and optical spectroscopes enable clumped isotope analysis of simple gaseous molecules like CO₂, O₂ and CH₄. These new clumping measurements have become powerful paleo-environmental proxies due to their unique sensitivity to the pathways and conditions of molecule formation, as well as their independence from bulk isotope ratios [1,2].

Clumped isotope measurements require the analysis of intact molecules, so gas-source instruments cannot be applied to dissolved species like oxyanions and organics. In ice core archives, the clumped isotopes of these soluble molecules are thought to contain key information that can reveal past oxidation pathways in Earth’s atmosphere. These insights may help unlock the elusive connection between climate and atmospheric chemistry, ultimately improving our ability to predict air quality and radiative forcing in a changing climate.

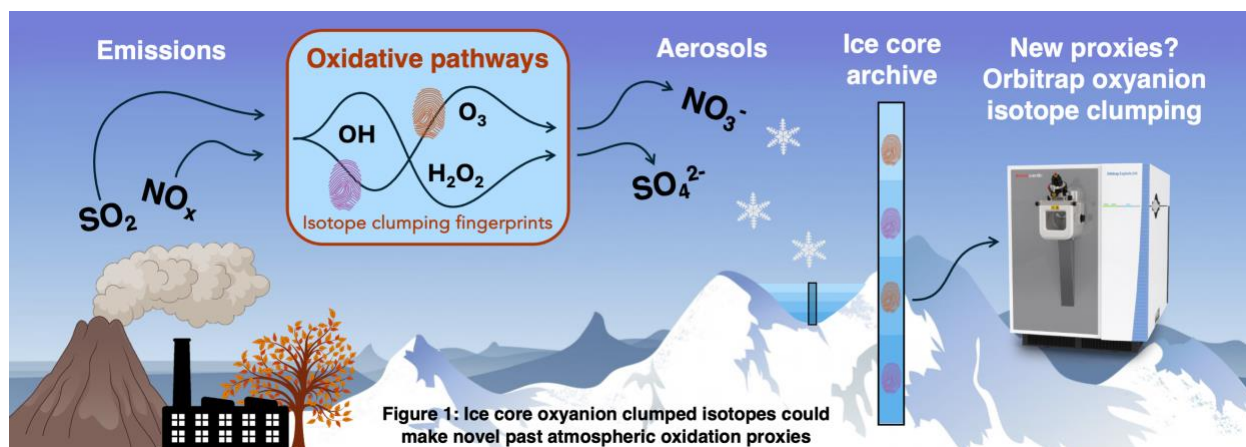
Recently, the Electrospray-Orbitrap mass spectrometer (ESI-Orbitrap MS) has been showcased as the upcoming new tool for stable isotopologue and clumped isotope analyses of intact soluble molecules. Its soft-ionisation and high mass-resolution enable, for the first time, simultaneous quantification of an extended range of singly and doubly substituted isotopologues of NO₃⁻ or SO₄²⁻ using just tens of nanomoles of sample – an improvement of 1-2 orders of magnitude [3,4,5].

Despite its impressive potential, questions remain regarding the accuracy, precision and robustness of ESI-Orbitrap MS measurements compared to the established IRMS and ICP-MS methods for NO₃⁻ and SO₄²⁻ stable isotope ratios in natural samples. To quantify its performance, we present a thorough intercomparison between ESI-Orbitrap MS and traditional bulk isotope measurements of the same ice core and aerosol samples.

Additionally, we present brand-new isotope clumping measurements of atmospheric NO₃⁻ and SO₄²⁻ archived in mid-to-low-latitude ice cores and invite suggestions towards the interpretation of these new palaeo-atmospheric proxies.

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- [2] Yeung *et al.* (2019) *Nature* <https://doi.org/10.1038/s41586-019-1277-1>.
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- [4] Neubauer *et al.* (2020) *Analytical Chemistry* <https://doi.org/10.1021/acs.analchem.9b04486>.
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TUESDAY

Oral Presentations

McIntosh et al, “Nitrogen isotope composition of amino acids in Bennu samples”

Samples collected from the carbonaceous B-type asteroid Bennu and delivered to Earth by the OSIRIS-REx mission provide a pristine record of the chemical processes that occurred in the early solar system [1]. Analysis of Bennu samples identified 33 amino acids with a combined abundance of ~70 nmol/g [2]. Additionally, ammonia, a potential precursor for the synthesis of amino acids and other soluble organic molecules, was detected in high abundance (~13.6 $\mu\text{mol/g}$) and enriched in ^{15}N ($+180 \pm 47\%$) [2]. The latter suggests that ammonia and possibly other nitrogen-containing soluble molecules formed in a cold molecular cloud or the outer regions of the protoplanetary disk. To determine whether amino acids from Bennu have an interstellar provenance, we measured their ^{15}N isotope values.

Sample OREX-800107-183 (0.2697 g) was homogenized into a powder following an established protocol [3]. The amino acid enantiomers were derivatized [after 4], producing N,O-bis(trifluoroacetyl) methyl esters. Commercial standards of amino acid enantiomers were used as a reference for the isotopic measurements.

Full separation of the derivatized amino acid enantiomers allowed peak trapping [after 5] and measurement of the nitrogen isotopic composition. Gas chromatography–electron impact ionization Orbitrap mass spectrometry [6] resulted in $\delta^{15}\text{N}$ values for glycine ($+185 \pm 10\%$) and b-alanine ($+170 \pm 4\%$).

The enriched ^{15}N values confirm the extraterrestrial provenance and suggest formation from the ^{15}N -enriched ammonia ($+180 \pm 47\%$) [2]. Likely formation pathways are Strecker synthesis for glycine and Michael addition for b-alanine. These ^{15}N -rich values point to a low-temperature origin permitting stable ammonia ice. Thus, Bennu's parent body, or components thereof, likely formed beyond Jupiter's orbit.

Supported by NASA under Contract NNM10AA11C and awards NNH21ZDA001N-ORSAPSP and NNH09ZDA007O.

References: [1] Laurretta D.S. & Connolly H.C. Jr. et al. (2024) *MAPS*, 59, 2453–2486. [2] Glavin D.P. & Dworkin J.P. et al. (2025) *Nat. Astron.*, 9, 199–210. [3] Elsilá J. E. et al. (2012) *MAPS*, 47(9), 1517–1536. [4] Zeichner S. et al. (2022) *Goldschmidt2022*, doi: 10.46427/gold2022.12538. [5] Eiler J. M. et al. (2017) *Int. J. Mass Spectrom.*, 422, 126–142. [6] McIntosh O. et al. (2025) *LPSC*, 1123.

Poster Presentations

Corbett et al., “Advancing Carboxylic Acid Quantification: A Novel Derivatisation Procedure for UHPLC-OT-MS with Applications to Asteroid Samples”

Recent sample return missions targeting asteroids Ryugu (Hayabusa2) and Bennu (OSIRIS-REx) have provided ample materials for the characterisation and quantification of soluble organic matter (SOM) in some of the most primitive objects in the solar system. While Bennu sample analyses have shown that monocarboxylic acids are present in relatively similar abundances as in the carbonaceous meteorites Murchison, Tarda, and Orgueil [1], hot water extractions of Ryugu samples have highlighted an apparent discrepancy in the abundance of carboxylic acids with chain lengths longer than that of formic or acetic acid [2]. This discrepancy has been interpreted as a result of increased aqueous alteration on Ryugu, but can potentially be attributed to analytical challenges, including low concentrations of certain carboxylic acids and matrix effects that limit sensitivity and reproducibility. In addressing these limitations, a new derivatisation procedure for ultra-high-performance liquid chromatography orbitrap mass spectrometry (UHPLC-OT-MS) aimed at improving detection of trace-level carboxylic acids in complex matrices has been developed. This protocol leverages enhanced ionisation efficiency and matrix cleanup techniques to reduce background interference and improve peak resolution, and expands the dynamic range for mono- and di-carboxylic acids, enabling the quantification of these molecules at low nanomolar concentrations. Preliminary testing using analogue matrices simulating asteroid conditions demonstrated the ability to consistently resolve low-abundance carboxylic acids offering new insights into organic synthesis, preservation, and degradation processes in extraterrestrial environments. The new protocol has significant implications for future analysis of returned asteroid samples, such as those from the aforementioned sample return missions. By closing the gap between meteorite and asteroid carboxylic acid profiles, this method can help refine theories on organic evolution in the early solar system, as well as improving our understanding of the chemical precursors to life. This work highlights the importance of combining advanced derivatisation strategies with high-resolution mass spectrometry to push the boundaries of analytical geochemistry.

[1] Glavin, D.P., Dworkin, J.P., Alexander, C.M.O. *et al.* Abundant ammonia and nitrogen-rich soluble organic matter in samples from asteroid (101955) Bennu. *Nat Astron* (2025).

[2] Potiszil, C. et al., Organic Matter in the Asteroid Ryugu: What We Know So Far. *Life* **13**, 1448 (2023).

Osorio et al., “Applying New Isotopic Approaches to Differentiate Sulfur Sources in Rainfall”

Transformations of sulfur (S) compounds in Earth’s atmosphere influence pollutant levels and the climate system. For example, reduced S emissions by natural ecosystems (e.g., dimethyl sulfide) and anthropogenic activity (e.g., sulfur dioxide) can be oxidized to sulfate, generating high-albedo aerosols and acidifying rainwater. Sulfate’s diverse sources and chemical pathways

complicate our understanding of the atmospheric S cycle, necessitating additional constraints to interpret the origin of atmospheric sulfate and its environmental consequences.

Houston, Texas presents a unique region to investigate differences in atmospheric S inputs to rainwater, including petrochemical infrastructure, farmlands, and seawater from the Gulf of Mexico. To elucidate their relative contributions to sulfate loading in rainwater, we studied the regional differences in rain chemistry at two sites: Rice University near downtown Houston (new sample collection for 8 months duration) and archived samples from the National Atmospheric Deposition Program (NADP) site (TX10) at the Attwater Prairie Chicken National Wildlife Refuge. Given the more remote, inland location of the NADP site, these samples are expected to have lower contributions of seawater and anthropogenic sources of S than the Rice University site. We found that major solute concentrations and elemental ratios at both sites indicate the presence of sea salt aerosols in precipitation from both areas. However, sea salt aerosols cannot explain total sulfate concentrations, implying the presence of non-sea-salt sulfate (NSS) in the form of anthropogenic and/or biogenic aerosols. Downtown measurements of NSS averaged ~82% of the total sulfate, while the NADP averaged ~86%; furthermore, absolute amounts of both total and NSS sulfate in the urban samples averaged 2.4-fold higher than the rural site. Together, these data suggest a greater influence of both sea-salt and anthropogenic sources on downtown Houston. Ongoing research includes collecting S and O isotopic data using electrospray ionization Orbitrap mass spectrometry, which could improve sulfate source tracing. This method requires roughly 1000 times less sulfate mass than traditional methods, which is important for dilute rain samples that would otherwise preclude isotopic analysis. Ultimately, this study pairs elemental analysis with novel isotopic method development to deconvolve controls on the atmospheric cycling of S.

Zang et al., “Experimental study on photochemical synthesis of building blocks from the CO/N₂O route on terrestrial planets”

Synthesis of building block molecules is crucial to understand the origins of life. Atmospheric UV photochemistry from CO₂ and CO is a plausible mechanism to generate organic compounds on terrestrial planets such as Earth and Mars [1]. In our previous study, it was demonstrated that solar UV irradiation to atmospheric N₂O under the presence of CO or H₂ with liquid water produce NH₃ as well as several simple amino acids [2]. Therefore, UV chemistry from CO/HCHO and N₂O/NH₃ is an alternative pathway to form amino acids or other building blocks without HCN. Here, we report further analysis of the experimental products systematically. After UV irradiation to CO, N₂O and H₂O, the samples were measured using HPLC-Orbitrap MS. The experiments yielded a variety of amino acids, imidazole, HMT and its derivatives, as well as some isomers of nucleobases. Among these products, HMT is the most abundant and formed by the reaction between HCHO and NH₃, which are produced from CO and N₂O, respectively. Furthermore, glycolaldehyde and glyoxal were determined and could initiate synthesis of further organic acids, sugars, and sugar acids. The amount of amino acids and glycolate increased largely when the products were hydrolyzed with 6M HCl at 110°C for 24 hours. Combining these results, amino acids as well as other building blocks were synthesized from HCHO and NH₃ through formose-type reaction, not via hydrogen cyanide (HCN) as an intermediate. Further

amino acid analysis and mechanisms will be discussed elsewhere in the conference (Asakura et al., in this volume). Based on the experimental results, the actual production rate of free glycine was estimated to be $1.2 \times 10^6 \sim 6.0 \times 10^7$ kg/yr when an atmosphere containing 1% of CO and 3000 ppbv N₂O. Our study demonstrated that the CO/N₂O route could be a plausible mechanism to provide significant amounts of building block molecules on early Earth or Mars.

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[2] Zang X., Ueno Y., and Kitadai N. (2022) Astrobiology, 22: 387-398.

WEDNESDAY

Oral Presentations

Meshoulam et al., “Carbon and hydrogen isotopic distributions in soluble polycyclic aromatic hydrocarbons in the Cold Bokkeveld meteorite”

Polycyclic aromatic hydrocarbons (PAHs) are abundant in nebulae, planetary atmospheres and meteorites, considered as one of the most abundant forms of C in the galaxy. These compounds can form at high temperatures (i.e., $T > 1000\text{K}$) in circumstellar environments, in cold molecular clouds and the interstellar medium ($\sim 10\text{--}50\text{K}$), via thermal and/or aqueous processes on carbonaceous planetesimals (100s K). Different reaction mechanisms have been suggested to dominate these different formation environments of PAHs. Therefore, constraining the mechanisms by which PAHs in meteorites were synthesized could inform understanding of the formation conditions and processes those meteorites have experienced.

In this work, we report the $\delta^{13}\text{C}$, δD and $\Delta^{2x13}\text{C}$ values of three PAHs (phenanthrene, fluoranthene and pyrene) extracted from the soluble organic matter of Cold Bokkeveld (CB) a carbonaceous chondrite. These meteorites are considered to be among the oldest materials in our solar system and therefore PAHs found on them may record the formation and evolution of organic matter in our solar system. Positive clumped isotopologue anomalies (i.e. $\Delta^{2x13}\text{C}$) of the three and four-ring PAHs extracted from Murchison and Ryugu¹ were previously interpreted as evidence that some fraction of PAHs on those bodies formed in cold ($< 50\text{K}$) environments.

The $\delta^{13}\text{C}_{\text{VPDB}}$ and $\delta\text{D}_{\text{VSMOW}}$ values of the CB PAHs are similar to the values measured for other CM chondrites and show similar relationships to molecular size and identity². However, unlike Murchison and Ryugu, in CB all the $\Delta^{2x13}\text{C}$ values of these three PAHs are $\approx 0\text{‰}$ providing no evidence that PAHs formed by cold chemistry mechanisms were preserved. We will discuss specific reaction mechanisms and conditions that could explain the measured $\delta^{13}\text{C}$, $\Delta^{2x13}\text{C}$ values.

References

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Poster Presentations

Girard et al., “Investigating nitrate sources and export in an alpine catchment using Orbitrap-based stable isotope techniques”

Alpine ecosystems are sensitive to subtle perturbations in reactive nitrogen (Nr) availability, with high rates of nitrogen deposition in these environments linked to species composition changes, biodiversity losses, acidification, and additional changes in soil chemistry. Atmospheric deposition of nitrate (NO_3^-) and ammonium (NH_4^+) is the sole mechanism by which anthropogenic N is transported to remote alpine watersheds, with winter deposition accumulated in the snowpack acting as a reservoir of Nr that is subsequently supplied to surface waters upon snowmelt. Stable isotope techniques have been used previously to investigate the relative contributions of NO_3^- deposition and soil nitrification to elevated surface water NO_3^- concentrations in alpine catchments. In this study, NO_3^- stable isotopic data ($\delta^{15}\text{N}$, $\delta^{18}\text{O}$, $\Delta^{17}\text{O}$) will be used to investigate the sources of NO_3^- in alpine lakes and lower elevation streams in the Sangre de Cristo Mountains of New Mexico (USA), the southernmost subrange of the Rocky Mountains (North America). While $\delta^{15}\text{N}$ and $\delta^{18}\text{O}$ data provide insight to the partitioning of anthropogenic nitrate sources (e.g., agricultural emissions, fossil fuel combustion), $\Delta^{17}\text{O}$ offers unique insight to the post-depositional processing of NO_3^- . This isotopic signature allows for exclusive differentiation between microbial NO_3^- production and atmospheric deposition. Atmospheric NO_3^- forms through the oxidation of nitrogen oxides (NO_x) by ozone (O_3) and has a positive $\Delta^{17}\text{O}$ value (20-35‰) that contrasts sharply with the signature of NO_3^- formed via nitrification ($\Delta^{17}\text{O} = 0\text{‰}$). Notably, the $\Delta^{17}\text{O}$ of atmospheric NO_3^- remains unaltered following deposition and subsequent environmental processing. To advance this research, we are developing a novel Orbitrap-based isotopic technique coupled with ion chromatography (IC), which stands to improve the throughput and accessibility of NO_3^- multi-isotopic tracer studies. This approach, alongside NO_3^- and NH_4^+ concentrations and isotopic data will be used to infer the sources and processes regulating NO_3^- export within this previously uncharacterized watershed. By presenting isotopic data from both conventional GC-IRMS and IC-Orbitrap-MS, we demonstrate the potential of Orbitrap-based stable isotope measurements for environmental applications.

THURSDAY

Oral Presentations

Monticelli et al., “Direct and simultaneous species-specific isotope ratio determination of inorganic and methyl mercury”

The isotopic composition of Hg is increasingly used to trace its sources and environmental pathways, with species-specific isotopic ratios providing essential insights into transformation processes such as methylation and demethylation. Distinguishing between the isotopic signatures of inorganic mercury (iHg) and the most abundant organic species, namely methylmercury (MeHg), is crucial for understanding their fate in the environment [1].

Current analytical methods for species-specific Hg isotope analysis rely primarily on chromatographic separations coupled to on- or off-line detection by multi-collector inductively coupled plasma mass spectrometry. While effective, these techniques involve time-consuming sample preparation, like derivatization and solvent extraction, as well as complex instrumental setups requiring the coupling of GC [2] or HPLC [3] to MC-ICPMS for on-line procedures. Additionally, the latter methods provide transient signals that may reduce the precision of the data.

We propose a novel procedure using soft ionization that preserves molecular integrity, enabling direct, species-specific Hg isotope ratio determination. The procedure features the complexation of inorganic Hg and methylmercury, its introduction into an ESI source, subsequent collisional dissociation of the complexes, followed by the direct simultaneous detection of inorganic mercury and MeHg isotopic patterns by an OrbitrapTM mass spectrometer. This approach circumvents derivatization and chromatographic steps, as it preserves molecular information, leading to simplified sample preparation and reduced processing time.

We will present the status of optimization and validation of the procedure and assess its accuracy, reproducibility, and applicability across different environmental matrices. We will also discuss the advantages and limitations compared to existing methods. It is also expected that this approach may be extended to the species specific detection of several elements and their isotopic ratios, offering a more efficient means of investigating metal species fractionation and cycling in natural and contaminated systems.

References

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[2] Suárez-Criado, Queipo-Abad, Rodríguez-González & García Alonso (2024), *Journal of Analytical Atomic Spectrometry* 39, 508-517.

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Roncoroni et al., “Simple Metal Ion Isotope Ratio Measurement with ESI-Orbitrap HRMS: Concept Validation and Preliminary Performance Evaluation”

Conventional methods for isotope ratio (IR) determination such as TIMS and MC-ICP-MS require extensive sample preparation [1] and struggle with isobaric interferences [2]. This study introduces a novel approach to convert a conventional Orbitrap™ MS into a platform for metal IR measurement, without any hardware modifications, allowing for a rapid and interference-free metal IR analysis.

The method exploits metal-ligand complexation to enhance ionization efficiency, followed by quadrupole filtering to isolate precursor ions. Subsequent collision-induced dissociation releases free metal ions for direct isotopic analysis with ultra-high resolution.

The isotopic patterns of several metal ions, namely Cr, Fe, Ni, Cu, Ga, In, Tl and Pb, were measured using EDTA as a complexant. The Orbitrap™ mass spectrometer effectively resolved isobaric interferences such as ^{54}Cr - ^{54}Fe , ^{58}Fe - ^{58}Ni , and ^{64}Ni - ^{64}Zn . Simultaneous multi-isotope and multi-element detection was achieved in a single scan, significantly improving throughput.

A proof-of-concept study on Pb isotopes was conducted, carefully evaluating the impact of chemical and instrumental parameters, to maximize precision and accuracy: under tuned conditions, the method demonstrated optimal accuracy within 2–14 ppm of reference values after mass bias correction. Precision was conversely in the range 0.11% - 0.023% as RSD% when measured over a long-term, 12 hours run. While precision is currently 10 to 50 times lower than TIMS and MC-ICP-MS, and comparable to single-collector ICP-MS, further optimization is undergoing to reduce inaccuracies in the measurements.

This method, though at an early stage, is a promising alternative for IR measurements that leverages the unique capabilities of Orbitrap™, offering a streamlined, interference-free approach to IR determination. Future work will focus on expanding the range of analyzable elements and suitable ligands, improving precision, with the potential to complement conventional techniques.

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[2] Van Acker, Theiner, Bolea-Fernandez, Vanhaecke & Koellensperger (2023), *Nat. Rev. Methods Prim.* 3, 1–18.

Beomsik Kim et al., “Isotopologue analysis of Polycyclic Aromatic Hydrocarbon (PAH) using APCI-Orbitrap MS and NMR”

Polycyclic aromatic hydrocarbons (PAHs) are widely distributed in the petroleum, soil, ocean, atmosphere, and extraterrestrial materials. Especially PAHs can be extracted directly from sediments and/or hydropyrolysis of sediment organic matter, including highly matured kerogen [1-2]. The isotopologues (^{13}C multiple and/or position-specific substituted) of PAHs may inform of modern and ancient carbon cycling, though stable isotope analysis of PAHs has been so far limited to bulk $^{13}\text{C}/^{12}\text{C}$ analysis through Gas Chromatography-Combustion-Isotope Ratio Mass Spectrometry (GC-C-IRMS) [3].

Here, we report method development of isotopologue analysis using high-mass-resolution Orbitrap mass spectrometer and Nuclear Magnetic Resonance (NMR). Commercial PAHs (anthracene, phenanthrene, pyrene, and fluoranthene) were introduced into the Orbitrap MS through Atmospheric Pressure Chemical Ionization (APCI). First, the mass species of each PAH—M0 (all ^{12}C), M1 (single ^{13}C substitution), M2 (double ^{13}C substitution), and M3 (triple ^{13}C substitution)—were measured simultaneously for ^{13}C clumped isotope analysis. The isotope ratios were calculated based on the intensity of each mass peak: $^{13}\text{x}1\text{R} = \text{I}_{\text{M1}}/\text{I}_{\text{M0}}$, $^{13}\text{x}2\text{R} = \text{I}_{\text{M2}}/\text{I}_{\text{M0}}$, and $^{13}\text{x}3\text{R} = \text{I}_{\text{M3}}/\text{I}_{\text{M0}}$. Then, δ values defined as $\delta^{13\text{x}n}\text{C} = (^{13\text{x}n}\text{R}_{\text{sample}}/^{13\text{x}n}\text{R}_{\text{standard}} - 1) \times 1000\text{‰}$, were calculated. We used dual syringe analysis of two different commercial PAHs as sample and standard, and calculate the clumped isotopologue abundance as $\Delta^{13\text{x}2}\text{C} = \delta^{13\text{x}2}\text{C} - 2\delta^{13\text{x}1}\text{C}$ and $\Delta^{13\text{x}3}\text{C} = \delta^{13\text{x}3}\text{C} - 3\delta^{13\text{x}1}\text{C}$, which represent relative difference of deviations from the stochastic distribution. Analytical precisions achieved range from 0.1‰ to 0.5‰ for $\delta^{13\text{x}1}\text{C}$, 0.06 to 1.3‰ for $\Delta^{13\text{x}2}\text{C}$, and 0.75‰ to 2.6‰ for $\Delta^{13\text{x}3}\text{C}$, when using a total sample amount of 17.5 nmol in 350 μL . In addition, position specific isotope analysis (PSIA) was made using the higher-energy collisional dissociation (HCD) cell. The ion fragments from each PAH were produced by the collision and introduced into the Orbitrap mass analyzer. In the case of phenanthrene ($\text{C}_{14}\text{H}_{10}$), C_4 to C_{14} fragments were identified, and their carbon isotope ratios were determined. These results were compared with independent analyses using ^{13}C -NMR. The combined isotopologue analysis using both Orbitrap MS and ^{13}C -NMR is useful to determine accurate clumped and position-specific isotopes within PAHs.

References

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- [2] Marshall et al. (2007), *Precambrian Research*, 155(1-2), 1-23.
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Dion-Kirschner et al., “Position-specific carbon isotope analysis of glucose and tree-ring cellulose by Orbitrap mass spectrometry”

The isotopic compositions of environmentally relevant compounds like glucose and cellulose carry traces of the metabolic processes they have undergone. However, many of the relevant fractionations are imparted at specific atomic positions, whereas most traditional isotope analyses measure the average isotopic composition of the whole molecule. Here, we present the first method for position-specific isotope analysis (PSIA) of glucose and cellulose via Electrospray-Ionization (ESI)-Orbitrap mass spectrometry. Unlike the few existing methods for glucose PSIA, our method enables measurement of natural isotope abundances with the submilligram sensitivity required for most geochemical applications.

Our new method produces carbon isotope ($\delta^{13}\text{C}$) measurements of 5 unique intramolecular sites within both reagent-grade and wood cellulose-derived glucose. We have determined optimal preparatory chemistry and instrument conditions, including the oxidation of glucose to gluconate prior to measurement. Our optimized method allows us to achieve precision of 0.7-2.4‰ for replicate measurements of positional $\delta^{13}\text{C}$ values. We further show that isotope measurements for each intramolecular fragment are accurate relative to externally verified $\delta^{13}\text{C}$ values within 0.1-0.6‰, typically outperforming instrument precision. We have applied this method to glucose standards from C3 and C4 plants, and find that the $\delta^{13}\text{C}$ values of intramolecular sites within both glucose standards vary by up to 12‰, but the relative positional trends are similar for both glucose sources. These measurements consume < 30 μg of glucose. We have further demonstrated accurate and precise measurements of cellulose-derived glucose from two tree species grown in climate chambers under varying conditions, with intramolecular $\delta^{13}\text{C}$ variability as high as 17‰. Our preliminary analyses, which consume < 1 mg of tree-ring cellulose, suggest that this variability records trees' ecophysiological response to water availability, likely with greater sensitivity than traditional compound-specific measurements. These findings highlight the promise of ^{13}C -PSIA via ESI-Orbitrap for gaining new insights into environmental carbon fluxes.

Hattori et al., “Triple oxygen isotope signature in methanesulfonate (MSA): method development and atmospheric application”

Dimethyl sulfide (DMS) is the primary natural sulfur source in the marine boundary layer, contributing to the formation of sulfate (SO_4^{2-}) and methanesulfonate (MSA, CH_3SO_3) through DMS chemistry. This process influences Earth's radiation budget, climate, and marine ecosystem by affecting atmospheric aerosol properties, which directly impact solar radiation and cloud properties. Understanding DMS chemistry is crucial for improving atmospheric models.

The mass-independent oxygen isotopic composition (^{17}O -excess, $\Delta^{17}\text{O}$) serves as a key tracer for atmospheric reactions. $\Delta^{17}\text{O}$ in Arctic ice cores reflects sulfate formation changes after SO_2 emission reductions in 1970 [1]. Recent Antarctic studies suggest MSA oxidation to SO_4^{2-} , with MSA potentially showing high $\Delta^{17}\text{O}$ values [2][3]. These findings impact the interpretation of

$\Delta^{17}\text{O}(\text{nss-SO}_4^{2-})$ in Antarctic ice cores, particularly regarding the glacial-to-interglacial $\Delta^{17}\text{O}$ shift [4], which needs reevaluation [3].

Our results indicate that MSA's $\Delta^{17}\text{O}$ could exceed 10‰, making it a potential tracer for the atmospheric DMS oxidation processes. However, traditional method using gas-source Isotope Ratio Mass Spectrometry (IRMS) are limited by sample size and analytical complexity. Here, we present a newly developed method using ESI-Orbitrap-MS to measure $\Delta^{17}\text{O}$ of MSA, achieving high sensitivity with a reproducibility of $\pm 1\text{‰}$ for 10 nmol MSA samples. This approach provides access to natural atmospheric and ice core samples, enhancing our ability to trace atmospheric DMS chemistry. This presentation also reports first observational results on $\Delta^{17}\text{O}$ of MSA in Antarctic aerosol samples.

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Poster Presentations

Wang & Hallmann, “Puzzle Game of Isotope Effects: Fingerprint and Reconstruct the Fates of Organic Molecules”

Stable isotope effects, as the key to unlock the code of kinetic, thermodynamic, biochemical, and even femto-chemical processes, have exhibited a power of blooming the breakthroughs in various areas of studies outside its origin in geology. However, its power gets limited by the current analytical techniques and theories, with which we hardly unravel a comprehensive observation of isotope effects in laboratory settings, simulations, or natural environments.

Our work aims to use the isotopic signatures of various environmental organics for environmental forensic and biogeochemistry studies. To this end we develop and validate novel stable isotope analysis methods, in view of advancing the next-generation mass spectrometry techniques (e.g., ESI-Orbitrap mass spectrometry), and establishing groundbreaking stable isotope effects theories.

Tracing the fate of various organic molecules (e.g., in soils, groundwaters, marine environments, fossils, and biomaterials) can be challenging because the precursors and/or products are either unknown/mixed, common in nature, or are labile and/or difficult to measure. Moreover, multiple transformation processes may occur simultaneously, rendering them difficult to distinguish or apportion based solely on tracking the concentrations. To resolve these challenges in environmental forensics, 2,4-dinitroanisole was used as a model compound for deciphering the transformation mechanisms using stable isotope effects, with the aim of more effective and efficient fate-tracking and source-identifications in field remediation efforts [1].

To explore broader applications in biogeochemistry, ESI-Orbitrap methods were developed and validated for a series of international standards (e.g., amino acids, urea), and in-house referencing materials (e.g., fatty acids), with nanomole-level analytes injected to achieve a precision for ^{13}C , ^{15}N , ^{18}O and ^{37}Cl of $< 0.2\%$. The capability for simultaneous high-precision analysis of multi-element isotope ratios in oxyanions and biomolecules is predicted to boost innovative discoveries and applications in geosciences and biochemistry. For example, in Earth's aqueous systems and metabolisms, reaction pathways may involve rate-determining steps of C-O or N-O bond cleavage or formation, causing relatively large primary oxygen kinetic isotope effects that could be diagnosed or clearly indicated by carbon and nitrogen isotope analysis.

References:

[1] C. Wang, M. E. Fuller, J. Murillo-Gelvez, et al. (2024) ES&T 58, 5996-6006.

Wojtal et al., “Stable Isotopic Analysis of Per- and Polyfluoroalkyl Substances (PFAS) by Nanoflow-HPLC-Orbitrap-MS”

Per- and polyfluoroalkyl substances (PFAS) are industrially derived fluorinated organic compounds that persist in the environment and are detrimental to human and ecosystem health. Recent studies have demonstrated stable isotopic differences in individual PFAS molecules like perfluorooctanoic acid (PFOA) and perfluorooctane sulfonic acid (PFOS) that might be exploited for source apportionment of contaminated environmental samples. Orbitrap mass spectrometry (Orbitrap-MS) is gaining popularity for isotope analysis. It presents a unique opportunity for the isotopic characterization of PFAS: 1) electrospray ionization (ESI) is a soft ionization technique thus intact molecules are introduced into the mass spectrometer 2) combustion of the molecules to simple gases is not required, and 3) significantly less material is required to make an isotopic measurement. Derivatization of PFAS molecules for isotopic analysis can be difficult and most PFAS do not combust using traditional isotope ratio mass spectrometry reactors. Additionally, Orbitrap mass spectrometry allows for the possibility of performing multi-element compound-specific isotope analysis, where the isotopic composition for a single molecule in multiple isotope systems is determined all at once. By combining multiple isotopic tools, we can separate different sources and processes in multi-dimensional space using a “fingerprinting” approach. We present results of a method developed for determining the isotopic compositions of PFOA, PFOS, PFBS, and HFPO-DA using HPLC-Orbitrap-MS and compare our results to traditional isotopic characterization of PFOA and PFOS using elemental analyzer-isotope ratio mass spectrometry (EA-IRMS) and thermal conversion elemental analyzer-isotope ratio mass spectrometry (TCEA-IRMS).

Kuhlbusch et al., “Workflow improvements for the Thermo Scientific™ Orbitrap Exploris™ Isotope Solutions”

Orbitrap-based Isotope Ratio MS is becoming increasingly visible in the community as a unique and complimentary approach to classical IRMS techniques for measuring relative abundances of isotopically substituted species. Electrospray ionization (ESI) offers the specific advantage of performing “soft” ionization, which produces intact molecular ions and provides unique insight into the molecular anatomy of polar compounds in aqueous solutions.

Drift associated with ESI ionization can be a major factor in data quality including the ultimate precision and accuracy achieved. Correcting for drift while measuring isotope ratios by ESI is done by sample-standard bracketing. This is currently achieved by one of two approaches. The dual syringe inlet approach uses a diverter valve switching between two syringes filled with sample or standard solution, enabling time efficient analysis. This method requires as little as 20 seconds of unusable time between samples and standards, allowing for close temporal measurement of a reference material. The major drawback of this approach is the necessity for the operator to be present to load the syringes. The second approach automizes the alternating injection of sample and standard solution using HPLC autosamplers. This approach enables fully automated analyses while lacking the temporal proximity of sample and standard seen with the dual syringe inlet. Cleaning and refilling of the autosampler loop causes a major gap (ca. minutes) of unused time between injections.

Here we will showcase a new workflow for isotope ratio standardization combining the best features from dual syringe inlet and autosampler injections. This method utilizes two flows of liquid controlled by a switching valve: one flow from an HPLC autosampler for sample and standard introduction, and a second flow is for a reference material. This setup allows for the injection of a reference material during the washout period of the HPLC, which reduces wasted time while simultaneously improving the quality of the measurement. This improved sample introduction technique enables efficient and fully automated sample analysis. Precision and accuracy are verified by drift correction and 1- or 2-point calibration. The novel approach will be demonstrated for inorganic (sulfate) and organic (vanillin) samples.

Andren et al., “ESI-Orbitrap-IRMS Analysis of Free Metabolites for Early Breast Cancer Detection and Insight into Metabolic Mechanisms”

Cancer cells rewire their metabolism as they proliferate, altering pathway activities and metabolite quantities. Isotopic analysis of metabolites in blood plasma could thus work as a non-invasive detection method and, in tissue, provide insight into the cancer metabolic network. Studies have shown cancerous cells to be ^{15}N -depleted and ^{13}C -enriched relative to adjacent non-cancerous cells[1]; however, primary isotopic anomalies occur at atomic sites undergoing bond-breaking and are not fully expressed in molecular average or bulk isotope analysis. With the advent of Orbitrap-IRMS, site-specific isotope analysis (SSIA) is now possible, which targets the atomic sites in the individual metabolites that experience the isotopic effect from the metabolically dysregulated reaction.

In this study, we focus on two metabolic hallmarks of breast cancer metabolism: the Warburg effect and glutamine addiction. We aim to determine whether SSIA of ^{13}C in free metabolites of Lactate and Alanine, and ^{15}N in free metabolites of Glutamine, and Arginine can serve as reliable biomarkers for breast cancer and whether specific isotopic patterns are unique for certain subtypes of breast cancer. Our approach involves three distinct stages: sample purification using reverse-phase HPLC and IC fraction collection, ESI-orbitrap-IRMS measurements, and data processing following the mathematical framework for isotomics. Preliminary quantification from the HPLC of these metabolites in the tissue samples reveals significant differences in cancerous versus adjacent healthy tissue, with cancerous tissues showing multi-fold higher concentrations, along with inter-cancer subtype variations. Using ESI-Orbitrap-IRMS, we achieve sub-permille accuracy on the molecular average for glutamine and approximately 1 permille accuracy for lactate on pure standards, verified by EA measurements. Future work will focus on sample purification and verifying the accuracy of SSIA before measuring biological samples. This study continues the work of SSIA on small quantities of polar organic molecules present in complex matrices. Reliable measurements of the intra-molecular isotopic fingerprints of these metabolites can provide valuable insight into the metabolic mechanisms that sustain cancer cell survival and can be used as a novel breast cancer diagnostic tool.

[1] Tea, I. *et al.* ^{13}C and ^{15}N natural isotope abundance reflects breast cancer cell metabolism. *Sci. Rep.* **6**, 34251 (2016)

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Nothing noted.